

Experiment #10

Implementation of First Come First Served and Shortest Job

First Scheduling Algorithm

Objective

In this lab we will learn about FCFS (First Come First Served) Scheduling Algorithm.

Software Tools

- ◆ Ubuntu 16.04
- ◆ GCC Compiler

Theory

FCFS:

CPU scheduler will decide which process should be given the CPU for its execution. For this it uses different algorithm to choose among the process. One among that algorithm is First Come First Serve (FCFS) algorithm. **First come first serve** (FCFS) scheduling algorithm simply schedules the jobs according to their arrival time. The process that enters the ready queue first is scheduled first, regardless of the size of its next CPU burst. The lesser the arrival time of the job, the sooner will the job get the CPU. FCFS scheduling may cause the problem of starvation if the burst time of the first process is the longest among all the jobs.

- Non-preemptive.
- Ready queue is a FIFO queue.
- Jobs arriving are placed at the end of queue.
- Dispatcher selects first job in queue and this job runs to completion of CPU burst.
- **Advantages:** simple, low overhead.
- **Disadvantage:** inappropriate for interactive systems.

Given n processes with their burst times and arrival times, the task is to find average waiting time and an average turnaround time using FCFS scheduling algorithm. FIFO simply queues processes in the order they arrive in the ready queue. Here, the process that comes first will be executed first and next process will start only after the previous gets fully executed.

Completion Time: Time at which the process completes its execution.

Turn Around Time: Time Difference between completion time and arrival time.

$$\text{Turn Around Time} = \text{Completion Time} - \text{Arrival Time}$$

Waiting Time (W.T): Time Difference between turnaround time and burst time.

$$\text{Waiting Time} = \text{Turn Around Time} - \text{Burst Time}.$$

Algorithm:

1. Input the processes along with their burst time (bt) and arrival time(at).
2. Find waiting time (wt) for all processes.
3. As first process that comes need not to wait so waiting time for process 1 will be 0 i.e. $wt[0] = 0$.
4. Find **waiting time** for all other processes i.e., for a given process i:

$$wt[i] = (bt[0] + bt[1] + \dots + bt[i-1]) - at[i].$$
5. Find **turnaround time** = waiting_time + burst_time for all processes.
6. Find **average waiting time** = total_waiting_time / no_of_processes.
7. Similarly, find **average turnaround time** = total_turn_around_time / no_of_processes.
8. Compute Average Waiting Time of processes.
9. Compute Average Turnaround Time of processes.
10. Display the Gantt chart.

SJF Algorithm

1. Input the processes along with their burst time (bt).
2. The process which have shortest burst will execute first i.e. sort processes according to their burst time.
1. If more than 1 process has same burst time then FCFS algorithm will be followed.
2. Find waiting time (wt) for all processes.
3. As first process that comes need not to wait so
 waiting time for process 1 will be 0 i.e. $wt[0] = 0$.
4. Find **waiting time** for all other processes i.e. for process i,

$$wt[i] = bt[i-1] + wt[i-1].$$
5. Find **turnaround time** = waiting_time + burst_time for all processes.
6. Find **average waiting time** = total_waiting_time / no_of_processes.

7. Similarly, find **average turnaround time** = total_turn_around_time / no_of_processes.
8. Compute Average Waiting Time of processes.
9. Compute Average Turnaround Time of processes.
10. Display the Gantt chart.

LAB TASK

1. Implement First Come First Serve CPU Scheduling Algorithm

OUTPUT 1

```
-----First Come First Serve-----
Enter the number of processes: 3
Enter 'Arrival Time' | 'Burst Time' for process p0: 0 24
Enter 'Arrival Time' | 'Burst Time' for process p1: 0 3
Enter 'Arrival Time' | 'Burst Time' for process p2: 0 3

      Process Name   Waiting Time   Turnaround Time
      p0
      p1
      p2

Average Waiting Time of Processes 17.000000
Average Turnaround Time of Processes 27.000000

-----GANTT CHART-----
| 0      p0      | 24      p1      | 27      p2      | 30
```

OUTPUT 2

```
-----First Come First Serve-----
Enter the number of processes: 4
Enter 'Arrival Time' | 'Burst Time' for process p0: 0 2
Enter 'Arrival Time' | 'Burst Time' for process p1: 2 5
Enter 'Arrival Time' | 'Burst Time' for process p2: 7 3
Enter 'Arrival Time' | 'Burst Time' for process p3: 8 10

      Process Name   Waiting Time   Turnaround Time
      p0
      p1
      p2
      p3

Average Waiting Time of Processes 0.500000
Average Turnaround Time of Processes 5.500000

-----GANTT CHART-----
| 0      p0      | 2      p1      | 7      p2      | 10      p3      | 20
```

2. Implement shortest Job First CPU Scheduling Algorithm.